Problem 3.A The Closest Pair

Given a set of points in a two dimensional space, you will have to find the distance between the closest two points.

Input

The input file contains several sets of input. Each set of input starts with an integer $N(0 \leq N \leq 10000)$, which denotes the number of points in this set. The next N line contains the coordinates of N two-dimensional points. The first of the two numbers denotes the X-coordinate and the latter denotes the Y-coordinate. The input is terminated by a set whose $N = 0$. This set should not be processed. The value of the coordinates will be less than 40000 and non-negative.

Output

For each set of input produce a single line of output containing a floating point number (with four digits after the decimal point) which denotes the distance between the closest two points. If there is no such two points in the input whose distance is less than 10000, print the line '`INFINITY`'.

Sample Input

```
3
0 0
10000 10000
20000 20000
5
0 2
6 67
43 71
39 107
189 140
0
```

Sample Output

```
INFINITY
36.2215
```

Problem 3.B Psychic Poker Player

In 5-card draw poker, a player is dealt a hand of five cards (which may be looked at). The player may then discard between zero and five of his or her cards and have them replaced by the same number of cards from the top of the deck (which is face down). The object is to maximize the value of the final hand. The different values of hands in poker are given at the end of this problem.

Normally the player cannot see the cards in the deck and so must use probability to decide which cards to discard. In this problem, we imagine that the poker player is psychic and knows which cards are on top of the deck. Write a program which advises the player which cards to discard so as to maximize the value of the resulting hand.

Input

Input will consist of a series of lines, each containing the initial five cards in the hand then the first five cards on top of the deck. Each card is represented as a two-character code. The first character is the face-value (`A`=Ace, `2`–`9`, `T`=10, `J`=Jack, `Q`=Queen, `K`=King) and the second character is the suit (`C`=Clubs, `D`=Diamonds, `H`=Hearts, `S`=Spades). Cards will be separated by single spaces. Each input line will be from a single valid deck, that is there will be no duplicate cards in each hand and deck.

Output

Each line of input should produce one line of output, consisting of the initial hand, the top five cards on the deck, and the best value of hand that is possible. Input is terminated by end of file.

Note: Use the sample input and output as a guide.

Note that the order of the cards in the player's hand is irrelevant, but the order of the cards in the deck is important because the discarded cards must be replaced from the top of the deck. Also note that examples of all types of hands appear in the sample output, with the hands shown in decreasing order of value.

Sample Input

```
TH JH QC QD QS QH KH AH 2S 6S
2H 2S 3H 3S 3C 2D 3D 6C 9C TH
2H 2S 3H 3S 3C 2D 9C 3D 6C TH
2H AD 5H AC 7H AH 6H 9H 4H 3C
AC 2D 9C 3S KD 5S 4D KS AS 4C
KS AH 2H 3C 4H KC 2C TC 2D AS
AH 2C 9S AD 3C QH KS JS JD KD
6C 9C 8C 2D 7C 2H TC 4C 9S AH
3D 5S 2H QD TD 6S KH 9H AD QH
```

Sample Output

```
Hand: TH JH QC QD QS Deck: QH KH AH 2S 6S Best hand: straight-flush
Hand: 2H 2S 3H 3S 3C Deck: 2D 3D 6C 9C TH Best hand: four-of-a-kind
Hand: 2H 2S 3H 3S 3C Deck: 2D 9C 3D 6C TH Best hand: full-house
Hand: 2H AD 5H AC 7H Deck: AH 6H 9H 4H 3C Best hand: flush
Hand: AC 2D 9C 3S KD Deck: 5S 4D KS AS 4C Best hand: straight
Hand: KS AH 2H 3C 4H Deck: KC 2C TC 2D AS Best hand: three-of-a-kind
Hand: AH 2C 9S AD 3C Deck: QH KS JS JD KD Best hand: two-pairs
Hand: 6C 9C 8C 2D 7C Deck: 2H TC 4C 9S AH Best hand: one-pair
Hand: 3D 5S 2H QD TD Deck: 6S KH 9H AD QH Best hand: highest-card
```

Problem 3.C Compound Words

You are to find all the two-word compound words in a dictionary. A two-word compound word is a word in the dictionary that is the concatenation of exactly two other words in the dictionary.

Input

Standard input consists of a number of lowercase words, one per line, in alphabetical order. There will be no more than 120,000 words.

Output

Your output should contain all the compound words, one per line, in alphabetical order.

Sample Input

```
a
alien
born
less
lien
never
nevertheless
new
newborn
the
zebra
```

Sample Output

```
alien
newborn
```

Caution time limit 3 sec.

Problem 3.D Oil Deposits

The GeoSurvComp geologic survey company is responsible for detecting underground oil deposits. GeoSurvComp works with one large rectangular region of land at a time, and creates a grid that divides the land into numerous square plots. It then analyzes each plot separately, using sensing equipment to determine whether or not the plot contains oil.

A plot containing oil is called a pocket. If two pockets are adjacent, then they are part of the same oil deposit. Oil deposits can be quite large and may contain numerous pockets. Your job is to determine how many different oil deposits are contained in a grid.

Input

The input file contains one or more grids. Each grid begins with a line containing m and n, the number of rows and columns in the grid, separated by a single space. If $m = 0$ it signals the end of the input; otherwise $1 \leq m \leq 100$ and $1 \leq n \leq 100$. Following this are m lines of n characters each (not counting the end-of-line characters). Each character corresponds to one plot, and is either '`*`', representing the absence of oil, or '`@`', representing an oil pocket.

Output

For each grid, output the number of distinct oil deposits. Two different pockets are part of the same oil deposit if they are adjacent horizontally, vertically, or diagonally. An oil deposit will not contain more than 100 pockets.

Sample Input

```
1 1
*
3 5
*@*@*
**@**
*@*@*
1 8
@@****@*
5 5
****@
*@@*@
*@**@
@@@*@
@@**@
0 0
```

Sample Output

```
0
1
2
2
```